

PDU Details

A detailed explanation of each of the supported PDU types.

As per the [SMPP 3.4 specification](#), each PDU has a mandatory header section that is the same for each PDU transmitted. These will not be outlined below (except for the *command_status* parameter in the *submit_sm_resp* PDUs).

Only parameters from the PDU body (mandatory parameters) and optional parameters are explained.

NOTE

If an optional parameter is not mentioned below, then it is not supported/is ignored.

PDU Types

The following PDU types and their responses are supported:

Command ID	Value (hex)	Description
<i>bind_receiver</i>	0x00000001	Establish a receiver bind.
<i>bind_transmitter</i>	0x00000002	Establish a transmitter bind.
<i>bind_transceiver</i>	0x00000009	Establish a transceiver bind.
<i>submit_sm</i>	0x00000004	Submit message to the SMSC for onward delivery to mobile device.
<i>deliver_sm</i>	0x00000005	Send message to ESME from SMSC (typically delivery receipt or MO message).
<i>unbind</i>	0x00000006	Unbind from SMSC and close SMPP session.
<i>enquire_link</i>	0x00000015	Initiate a check to confirm ESME/SMSC is still reachable.
<i>generic_nack</i>	0x80000000	Acknowledge unrecognized or corrupt PDU.

NOTE

data_sm, *query_sm*, *cancel_sm*, *replace_sm* and *submit_sm_multi* are **not** supported.

Bind PDU

An SMPP *bind_receiver*, *bind_transmitter* or *bind_transceiver* PDU request has a fixed set of fields many of which are irrelevant to us.

Name	Description
<i>system_id</i>	Your SMPP username (maximum of 15 characters allowed)
<i>password</i>	Your SMPP password (maximum of 8 characters allowed)
<i>system_type</i>	Ignored with the exception of "test" which is no longer supported and will be rejected.
<i>interface_version</i>	The SMPP version that you want to use. Only version 3.4 (0x34) is supported.
<i>addr_ton</i>	Ignored
<i>addr_npi</i>	Ignored
<i>address_range</i>	Ignored

enquire_link

To avoid being disconnected, you need to send at least one *enquire_link* PDU on every successfully established

connection (bound session) every 2 minutes.

It's recommended that you send the *enquire_link* PDU once every 30 seconds to timeously detect any lost connections. SMSPortal will also send an *enquire_link* every 30 seconds to detect any lost connections.

submit_sm

You can use the *submit_sm* PDU to send us your MT messages. Like a bind request, the *submit_sm* PDU request also has a couple of fields that are unused by our platform and can safely be ignored.

Name	Description
<i>service_type</i>	Ignored
<i>source_addr_ton</i>	Ignored but needs to be a valid value as per the SMPP specification
<i>source_addr_npi</i>	Ignored but needs to be a valid value as per the SMPP specification
<i>source_addr</i>	The sender ID or valid two-way number. If the sender ID is not valid on the network, this will be replaced by SMSPortal using a valid Sender ID. The original <i>source_addr</i> you provided will still be returned in the <i>deliver_sm</i> for matching purposes.
<i>dest_addr_ton</i>	Ignored but needs to be a valid value as per the SMPP specification
<i>dest_addr_npi</i>	Ignored but needs to be a valid value as per the SMPP specification
<i>destination_addr</i>	The mobile number of the handset to which the message must be delivered (MSISDN). Note: The MSISDN needs to be in international numbering format (E.164) without any symbols (e.g. plus sign) or it will be rejected with an ESME_RINVDSTADR error response.
<i>esm_class</i>	Message mode and type. Refer to the SMPP specification for more information.
<i>protocol_id</i>	Ignored
<i>priority_flag</i>	Ignored
<i>schedule_delivery_time</i>	Not supported - Any messages with a scheduled time are overridden and sent out for immediate delivery.
<i>validity_period</i>	The validity period of the message. If the first attempt fails, the mobile network operator will make several delivery attempts during this timeframe. This is usually specified using the absolute time format but relative time format is also valid. SMSPortal's default <i>validity_period</i> is 48 hours if no value is provided. For One-Time Passwords, we recommend setting the validity to a shorter period (e.g. 1 hour). This will make the message expire when the content is no longer relevant and potentially force the mobile network operator to perform more retries within a short time frame after sending the message, hence increasing the chance of delivery.
<i>registered_delivery</i>	Set to 1 to receive delivery receipts when delivery was successful or failed. Set to 2 to receive delivery receipts only when delivery failed. All other values are ignored (i.e. no delivery receipt will be generated).
<i>replace_if_present_flag</i>	Ignored
<i>data_coding</i>	SMSC default character set is GSM7 See Data Coding Schemes for further information.
<i>sm_default_msg_id</i>	Ignored
<i>sm_length</i>	The length of the <i>short_message</i> field. Valid values are 0 – 254.
<i>short_message</i>	The contents of the message to be sent.
<i>sarmsg_ref_num</i> <i>sar_total_segments</i> <i>sar_segment_seqnum</i>	Optional parameters; see SMPP specification on how they should be constructed. The sar parameters can be used to identify concatenated message parts instead of encoding UDH in the <i>short_message</i> field.
<i>Customer Reference</i>	Vendor Specific TLV Type 0x1400 Can be used when generating reports using the control panel.

Name	Description
<i>Cost Center Reference</i>	Vendor Specific TLV Type 0x1401 Can be used when generating reports using the control panel.

submit_sm_resp

Name	Description
<i>command_status</i>	As per the SMPP specification
<i>message_id</i>	The <i>message_id</i> will always be returned as a 64 (or less) character string as per the SMPP specification .

deliver_sm for delivery receipts (DLR)

DLRs will be sent to you via a *deliver_sm* PDU. The fields are exactly the same as a *submit_sm* PDU, but there will be differences in which fields you are free to ignore, and which you are not.

Name	Description
<i>service_type</i>	Always NULL, can be ignored
<i>source_addr_ton</i>	Autodetected value based on the value returned by the Mobile Network Operator
<i>source_addr_npi</i>	Autodetected value based on the value returned by the Mobile Network Operator
<i>source_addr</i>	The value that you specified in the <i>destination_addr</i> field of the original <i>submit_sm</i> PDU.
<i>dest_addr_ton</i>	Autodetected value based on the value returned by the Mobile Network Operator
<i>dest_addr_npi</i>	Autodetected value based on the value returned by the Mobile Network Operator
<i>destination_addr</i>	The value that you specified in the <i>source_addr</i> field of the original <i>submit_sm</i> PDU.
<i>esm_class</i>	Usually 4 (can be used to differentiate between a DLR and MO message). Refer to the SMPP specification for more information.
<i>protocol_id</i>	Always 0, can be ignored.
<i>priority_flag</i>	Always 0, can be ignored.
<i>schedule_delivery_time</i>	Always NULL, can be ignored.
<i>validity_period</i>	Always NULL, can be ignored.
<i>registered_delivery</i>	Always 0, can be ignored.
<i>replace_if_present_flag</i>	Always 0, can be ignored.
<i>data_coding</i>	Always 0, the SMSC always uses the GSM alphabet in delivery receipts.
<i>sm_default_msg_id</i>	Always 0, can be ignored.
<i>sm_length</i>	The length of the <i>short_message</i> field.
<i>short_message</i>	Informational content of the delivery receipt, see Delivery receipt short message format below.
<i>receipted_message_id</i>	Optional Parameter: The same <i>messageid</i> that was returned in the <i>submit_sm_resp</i> PDU when the message was accepted.

IMPORTANT

If you are not bound with a receiver or transceiver, the SMSC will queue delivery receipts for a maximum of 12 hours or until a maximum queue size of 1 million PDUs has been reached, after which the oldest *deliver_sm* PDUs will be discarded to make room for newer *deliver_sm* PDUs.

deliver_sm for MO messages

MOs will also be sent to you via a *deliver_sm* PDU. The fields are exactly the same as a *submit_sm* PDU, but there will be differences in which fields you are free to ignore, and which you are not.

Name	Description
<i>service_type</i>	Always NULL, can be ignored
<i>source_addr_ton</i>	Autodetected and set according to the value of <i>source_addr</i> .
<i>source_addr_npi</i>	Autodetected and set according to the value of <i>source_addr</i> .
<i>source_addr</i>	The sender of the MO
<i>dest_addr_ton</i>	Autodetected and set according to the value of <i>destination_addr</i> .
<i>dest_addr_npi</i>	Autodetected and set according to the value of <i>destination_addr</i> .
<i>destination_addr</i>	The value that you specified in the <i>source_addr</i> field of the original <i>submit_sm</i> PDU. On request, this setting can be changed to return the value that the mobile network operator provided in the <i>source_addr</i> field.
<i>esm_class</i>	Typically 0 or 64. A value of 64 (UDHI) indicates the presence of a user data header in the <i>short_message</i> field. Refer to the SMPP specification for more information.
<i>protocol_id</i>	Always 0, can be ignored.
<i>priority_flag</i>	Always 0, can be ignored.
<i>schedule_delivery_time</i>	Always NULL, can be ignored.
<i>validity_period</i>	Always NULL, can be ignored.
<i>registered_delivery</i>	Always 0, can be ignored.
<i>replace_if_present_flag</i>	Always 0, can be ignored.
<i>data_coding</i>	Will be 0 for plain-text message content which will be in the SMSC's default character set of GSM as per https://en.wikipedia.org/wiki/GSM_03.38 Will be 8 for Unicode message content which will be in the UCS-2 character set.
<i>sm_default_msg_id</i>	Always 0, can be ignored.
<i>sm_length</i>	The length of the <i>short_message</i> field.
<i>short_message</i>	The MO message content.
<i>receipted_message_id</i>	Optional Parameter: The same <i>message_id</i> that was returned in the <i>submit_sm_resp</i> PDU when the message was accepted.

IMPORTANT

If you are not bound with a receiver or transceiver, the SMSC will queue MO messages for a maximum of 12 hours or until a maximum queue size of 1 million PDUs has been reached, after which the oldest

deliver_sm PDUs will be discarded to make room for newer *deliver_sm* PDUs.

Delivery receipt short_message format

The *short_message* parameter in the *deliver_sm* PDU for a delivery receipt will contain report information regarding the delivery result of the message. The standard format as per the [SMPP specification](#) is used:

Example

```
id:IIIIIIIIII sub:SSS dlvrD:DDD submit date:YYMMDDhhmm done date:YYMMDDhhmm stat:DDDDDD err:EEE
*id:03/199440/UKQKF/5XrwwB/00004 sub:000 dlvrD:000 submit date:2111152111 done date:2111152111 st
```

Name	Description
<i>id</i>	The same <i>message_id</i> that was returned in the <i>submit_sm_resp</i> when the message was accepted.
<i>sub</i>	Always set to 0
<i>dlvrD</i>	Always set to 0
<i>submit date</i>	The date and time when the ESME submitted the message to the SMSC.
<i>done date</i>	The date and time when the SMSC received the specified status.
<i>stat</i>	DELIVRD, UNDELIV, BLIST, ACCEPTD or REJECTD. See Delivery Statuses for more information.
<i>err</i>	Always 000 for successful and failed delivery. The only exception is 001 which indicates that number is on the DNC list and that SMS delivery is restricted due to a request by the owner or WASPA.
<i>text</i>	The first 20 characters of the original <i>submit_sm</i> as per the SMPP specification .

deliver_sm_resp

Name	Description
<i>message_id</i>	Ignored

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